

Мета: отримати базові навички по використанню AWS Management Console та створити власний мікро-сервер для подальшого використання

Завдання:

- Реєстрація в AWS
- Створення власного віртуального мікро-сервера
- Отримання віддаленого доступу через SSH
- Вивчення елементів моніторингу серверу та налаштування
- Ознайомлення з IAM сервісом для керування доступами
- Документування зробленої роботи у вигляді деталізованого протоколу з коментарями

1. Реєстрація в AWS

Створила акаунт

Account

Account info

Close account

Account details

Actions

Name	sonkkasonkka	Service provider	Amazon Web Services EMEA SARL
ID	418862088861	ARN	arn:aws:account::418862088861:account

Account display settings - new

Edit

Account color

Unset

Contact information info

Edit

Full name	Кувавiна Софiя Олексiївна	Company name	None	Address	КПІ
Phone number	+380 664101967	Website URL	None		Київ, Kyiv 03056
					UA

2. Створення інстансу

У сервісі EC2 створила новий інстанс

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

labka1

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

[AMI from catalog](#)

[Recents](#)

[Quick Start](#)

Name

amzn2-ami-hvm-2.0.20260120.1-x86_64-gp2

Verified provider

Free tier eligible

Description

Amazon Linux 2 AMI 2.0.20260120.1 x86_64 HVM gp2

Image ID

ami-026992d753d5622bc

Username

ec2-user



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2 AMI (HVM), SSD Volume Type (64-bit x86) Operating System
ami-026992d753d5622bc

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#)

[Launch instance](#)

[Preview code](#)

Роблю ключі для openssh у pem-форматі

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

pair1

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

pair1.pem	02.02.2026 19:40	Файл "PEM"	2 КБ
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Налаштування мережових параметрів

Під час налаштування мережі є дозвіл на SSH-з'єднання, встановивши параметр:
Allow SSH traffic from Anywhere

▼ Network settings

Info

Edit

Network

Info

vpc-0fd04462f9b82d0a0

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'Amazon Linux 2 AMI (HVM), SSD Volume Type (64-bit x86) Operating System-2.0.20260120.1-AutogenByAWSMP--1' with the following rules:

☒ Allow SSH traffic from

Recommended rule from AMI

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

Налаштування сховища instance

▼ Configure storage

Info

Advanced

1x

8

GiB

gp3

Root volume, 3000 IOPS, Not encrypted

Add new volume

🕒 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

↻

0 x File systems

Edit

Запускаємо наш instance

C2 > Instances > i-070346724e9ef7da7

🔍

board

lobal View

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nices

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ice Types

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requests

js Plans

ved Instances

ated Hosts

ity Reservations

ity Manager

es

atalog

ic Block Store

yes

horts

Instance summary for i-070346724e9ef7da7 (labka1)

Updated less than a minute ago

Instance ID

i-070346724e9ef7da7

IPv6 address

-

Hostname type

IP name: ip-172-31-64-139.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

3.235.172.119 [Public IP]

IAM Role

-

IMDSv2

Optional

⚠ EC2 recommends setting IMDSv2 to required | Learn more

Operator

-

Public IPv4 address

3.235.172.119 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-64-139.ec2.internal

Instance type

t3.micro

VPC ID

vpc-0fd04462f9b82d0a0

Subnet ID

subnet-08b06fe07b57c84f6

Instance ARN

arn:aws:ec2:us-east-1:418862088861:instance/i-070346724e9ef7da7

Private IPv4 addresses

172.31.64.139

Public DNS

ec2-3-235-172-119.compute-1.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Connect

Instance state

Actions

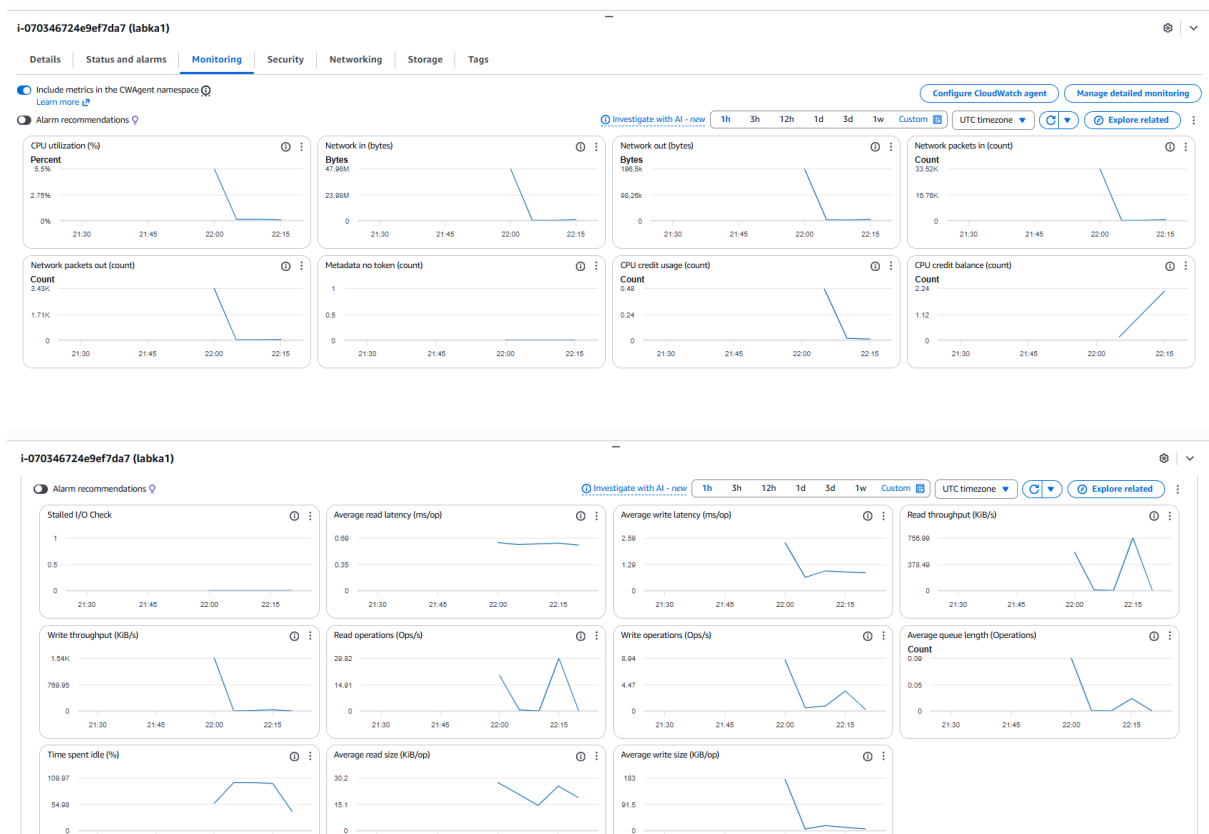
3. Підключення до інстансу

Знаходимо публічну ip-адресу інстансу



4. Навчитись моніторити використання ресурсів засобами Linux та веб-консолі керування інстансами

Я переглянула доступні інструменти AWS для моніторингу роботи інстансу, де можна бачити завантаження процесора, вхідний та вихідний трафік та інші показники. Все це можна зручно налаштовувати и реагувати на будь-які помилки



Більш детально моніторинг ресурсів я виконала за допомогою сервісу Amazon CloudWatch. На графіку відображено метрику CPUUtilization, яка показує рівень

The screenshot displays the AWS CloudWatch console for a Lambda function. The 'Metrics' tab is selected, showing a line graph for 'CPUUtilization'. The y-axis represents 'Percent' from 0% to 100%, and the x-axis shows time from 19:30 to 22:30. A sharp spike in CPU utilization is visible at 22:00, reaching 100%. Below the graph, a table lists metrics for the function 'labka1'. The 'CPUUtilization' metric is highlighted, showing a value of 100% at 22:00.

Metric	Value	Unit	Alarm
labka1	100	%	No alarms

Моніторинг завантаження CPU і процесів (top)

```

bash: echo: command not found
[ec2-user@ip-172-31-64-139 ~]$ top
top - 22:25:24 up 8 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 86 total, 1 running, 47 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 962712 total, 683688 free, 72868 used, 206156 buff/cache
KiB Swap: 0 total, 0 free, 0 used. 752964 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   41456   5284   3868  S   0.0   0.5   0:01.60 systemd
    2 root        20   0         0      0      0  S   0.0   0.0   0:00.00 kthreadd
    4 root         0 -20         0      0      0  I   0.0   0.0   0:00.00 kworker/0:0H
    6 root         0 -20         0      0      0  I   0.0   0.0   0:00.00 mm_percpu_wq
    7 root        20   0         0      0      0  S   0.0   0.0   0:00.03 ksoftirqd/0
    8 root        20   0         0      0      0  I   0.0   0.0   0:00.11 rcu_sched
    9 root        20   0         0      0      0  I   0.0   0.0   0:00.00 rcu_bh
   10 root        rt    0         0      0      0  S   0.0   0.0   0:00.00 migration/0
   11 root        rt    0         0      0      0  S   0.0   0.0   0:00.00 watchdog/0
   12 root        20   0         0      0      0  S   0.0   0.0   0:00.00 cpuhp/0
   13 root        20   0         0      0      0  S   0.0   0.0   0:00.00 cpuhp/1
   14 root        rt    0         0      0      0  S   0.0   0.0   0:00.00 watchdog/1
   15 root        rt    0         0      0      0  S   0.0   0.0   0:00.21 migration/1
   16 root        20   0         0      0      0  S   0.0   0.0   0:00.02 ksoftirqd/1
   17 root        20   0         0      0      0  I   0.0   0.0   0:00.05 kworker/1:0
   18 root         0 -20         0      0      0  I   0.0   0.0   0:00.00 kworker/1:0H
   20 root        20   0         0      0      0  S   0.0   0.0   0:00.00 kdevtmpfs
   21 root         0 -20         0      0      0  I   0.0   0.0   0:00.00 netns
   22 root        20   0         0      0      0  I   0.0   0.0   0:00.02 kworker/u4:1
   35 root        20   0         0      0      0  I   0.0   0.0   0:00.01 kworker/0:1
  106 root        20   0         0      0      0  S   0.0   0.0   0:00.00 khungtaskd
  134 root        20   0         0      0      0  S   0.0   0.0   0:00.00 oom_reaper
  158 root         0 -20         0      0      0  I   0.0   0.0   0:00.00 writeback
  182 root        20   0         0      0      0  S   0.0   0.0   0:00.00 kcompactd0

```

```
[ec2-user@ip-172-31-64-139 ~]$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	940M	70M	668M	420K	201M	735M
Swap:	0B	0B	0B			

Загрузка диска (df -h)

```
Swap: 0B 0B 0B
[ec2-user@ip-172-31-64-139 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         460M    0  460M   0% /dev
tmpfs            471M    0  471M   0% /dev/shm
tmpfs            471M  420K  470M   1% /run
tmpfs            471M    0  471M   0% /sys/fs/cgroup
/dev/nvme0n1p1   8.0G  1.7G  6.3G  22% /
tmpfs            95M    0   95M   0% /run/user/1000
[ec2-user@ip-172-31-64-139 ~]$
```

Моніторинг загрузки диску (iostat -dx 1)

```
[ec2-user@ip-172-31-64-139 ~]$ iostat -dx 1
Linux 4.14.355-280.713.amzn2.x86_64 (ip-172-31-64-139.ec2.internal) 02/03/2026 _x86_64_ (2 C
PU)
Device: svctm %util rrqm/s wrqm/s r/s w/s rkB/s wkB/s avgrq-sz avgqu-sz await r_await w_await
nvme0n1 0.09 0.13 0.00 0.05 12.68 1.43 343.55 17.01 51.12 0.00 0.67 0.66 0.75
Device: svctm %util rrqm/s wrqm/s r/s w/s rkB/s wkB/s avgrq-sz avgqu-sz await r_await w_await
nvme0n1 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Device: svctm %util rrqm/s wrqm/s r/s w/s rkB/s wkB/s avgrq-sz avgqu-sz await r_await w_await
nvme0n1 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Device: svctm %util rrqm/s wrqm/s r/s w/s rkB/s wkB/s avgrq-sz avgqu-sz await r_await w_await
nvme0n1 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Device: svctm %util rrqm/s wrqm/s r/s w/s rkB/s wkB/s avgrq-sz avgqu-sz await r_await w_await
nvme0n1 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Device: svctm %util rrqm/s wrqm/s r/s w/s rkB/s wkB/s avgrq-sz avgqu-sz await r_await w_await
nvme0n1 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Device: svctm %util rrqm/s wrqm/s r/s w/s rkB/s wkB/s avgrq-sz avgqu-sz await r_await w_await
nvme0n1 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
```

як довго працює сервер, загальна кількість користувачів(Uptime)

```
[ec2-user@ip-172-31-64-139 ~]$ uptime
22:27:07 up 10 min, 1 user, load average: 0.00, 0.00, 0.00
[ec2-user@ip-172-31-64-139 ~]$
```

статистика віртуальної пам'яті(Vmstat)

```
22:27:07 up 10 min, 1 user, load average: 0.00, 0.00, 0.00
[ec2-user@ip-172-31-64-139 ~]$ vmstat
procs -----memory----- --swap-- ----io---- -system-- -----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st
 1 0  0 683928 2088 204176 0 0 161 8 69 120 1 0 98 0 0
[ec2-user@ip-172-31-64-139 ~]$
```

мережева статистика(Netstat -s)

```
[ec2-user@ip-172-31-64-139 ~]$ netstat -s
Ip:
  Forwarding: 2
  1179 total packets received
  1 with invalid addresses
  0 forwarded
  0 incoming packets discarded
  1178 incoming packets delivered
  952 requests sent out
  61 dropped because of missing route
Icmp:
  0 ICMP messages received
  0 input ICMP message failed
  ICMP input histogram:
  0 ICMP messages sent
  0 ICMP messages failed
  ICMP output histogram:
Tcp:
  91 active connection openings
  1 passive connection openings
  0 failed connection attempts
  2 connection resets received
  1 connections established
  1053 segments received
  836 segments sent out
  4 segments retransmitted
  0 bad segments received
  6 resets sent
Udp:
  125 packets received
  0 packets to unknown port received
  0 packet receive errors
  125 packets sent
  0 receive buffer errors
  0 send buffer errors
UdpLite:
TcpExt:
  2 TCP sockets finished time wait in fast timer
  6 delayed acks sent
  Quick ack mode was activated 5 times
  472 packet headers predicted
  240 acknowledgments not containing data payload received
  70 predicted acknowledgments
  TCPTimeouts: 3
  TCPLossProbes: 1
```

У вкладці Security Groups можемо побачити вхідний та вихідний трафік.

sg-0a6030529cddc785b - launch-wizard-1

Actions

Details

Security group name

launch-wizard-1

Owner

418862088861

Security group ID

sg-0a6030529cddc785b

Inbound rules count

1 Permission entry

Description

launch-wizard-1 created 2026-02-02T17:36:06.869Z

VPC ID

vpc-0f0d4462f9b82d0a0

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
	-	sg-r0c748160626f5b4cb	IPv4	SSH	TCP	22	0.0.0.0/0	-

sg-086987a5aeb36c934 - default

Actions

Details

Security group name

default

Owner

418862088861

Security group ID

sg-086987a5aeb36c934

Inbound rules count

1 Permission entry

Description

default VPC security group

VPC ID

vpc-01d04462f9b82d0a0

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

Name

Security group rule ID

IP version

Type

Protocol

Port range

Source

Description

-

sgr-0f4e938f02bdba3d5

-

All traffic

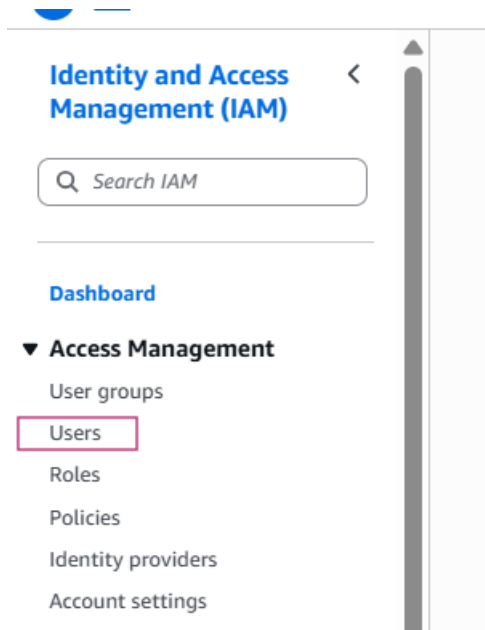
All

All

sg-086987a5aeb36c934...

-

Заходжу до IAM--Users



Та створюю нового користувача

Specify user details

User details
User name

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , _ @ - (hyphen)

☐ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

📘 If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.

[Learn more](#)

[Cancel](#) [Next](#)

Назначаю повний доступ до S3

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1443) [Create policy](#)
Choose one or more policies to attach to your new user.

[×](#)

Filter by Type

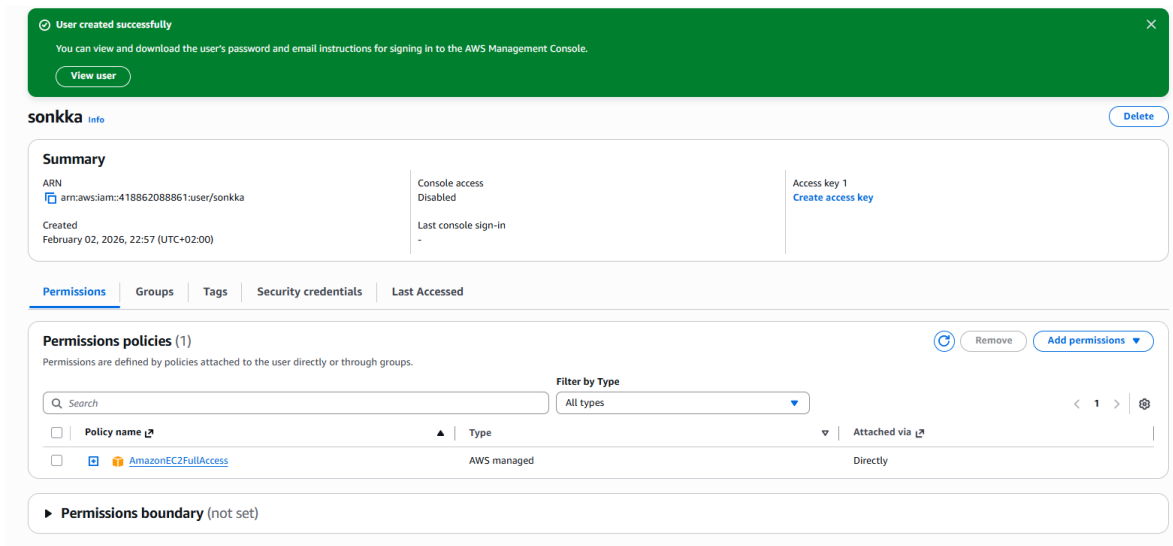
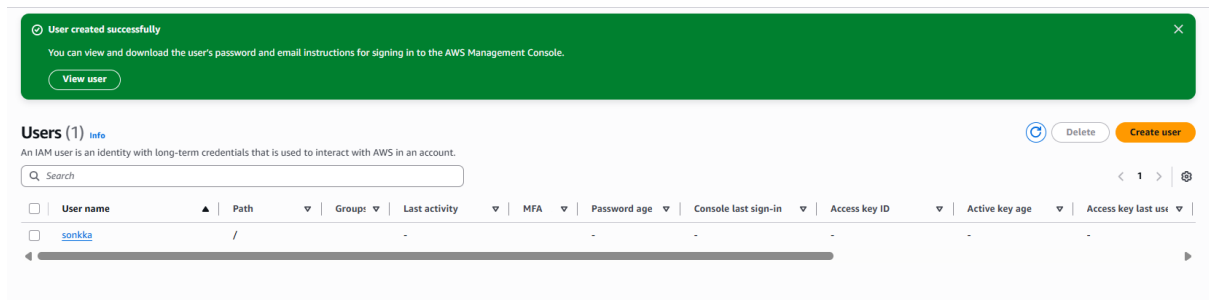
All types [▼](#) 1 match

☒	Policy name ↗	Type	Attached entities
☒	AmazonEC2FullAccess	AWS managed	0

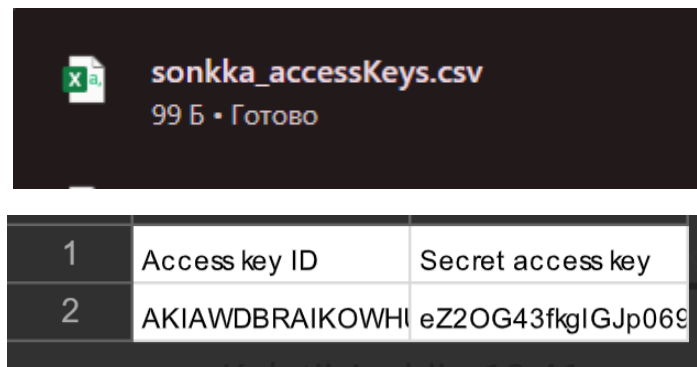
▶ Set permissions boundary - optional

[Cancel](#) [Previous](#) [Next](#)

Користувач створений!



В методичці бачимо: Після створення користувача, ви побачите інформацію про нього, а також можливість завантажити CSV-файл, який містить ключі доступу та секретні ключі в розділі Security Credentials.



Висновки

У ході виконання лабораторної роботи я отримала базові навички роботи з хмарною платформою AWS, створення та налаштування EC2 інстансу, підключення до сервера, моніторингу використання ресурсів та керування доступом за допомогою IAM.